Introduction

Fisher's exact test assesses the association between two categorical variables in a contingency table without relying on the chi-squared large-sample approximation. It is the gold standard for 2x2 tables with small cells, and extends to $r \times c$ tables via network algorithms.

Prerequisites

Contingency tables, hypergeometric distribution.

Theory

For a 2x2 table with observed counts and fixed marginal totals, the null distribution of the number of "exposed cases" conditional on the margins is **hypergeometric**:

$$P(X = k) = \frac{\binom{R_1}{k}\binom{R_2}{C_1-k}}{\binom{n}{C_1}}.$$

The exact two-sided p-value is the sum of probabilities of all tables as extreme or more extreme than the observed one (typically ranked by probability or by the odds ratio).

The test statistic implicitly reported is the **odds ratio**:

$$\widehat{OR} = \frac{ad}{bc}.$$

Fisher's test produces a p-value and an exact CI for the OR.

Assumptions

- Independent observations.
- 2x2 contingency table (extensions exist for larger tables).
- Fixed marginals under the null (conditional inference).

R Implementation

```r
# 2x2 table: small-cell case
tab <- matrix(c(3, 15,
                12, 20), nrow = 2, byrow = TRUE,
              dimnames = list(Treatment = c("Drug", "Placebo"),
                              Outcome   = c("Event", "No event")))
tab

fisher.test(tab)

# For comparison, chi-squared (warning expected due to small cells)
chisq.test(tab)

# Larger table: Fisher via network algorithm
big <- matrix(c(4, 2, 1,
                3, 8, 5,
                2, 3, 6), nrow = 3, byrow = TRUE)
fisher.test(big)
```

Output & Results

```
        Fisher's Exact Test for Count Data
data:  tab
p-value = 0.02676
alternative hypothesis: true odds ratio is not equal to 1
95 percent confidence interval:
 0.027 0.918
sample estimates:
odds ratio: 0.179
```

The exact p-value is 0.027; OR = 0.18 (95 % CI 0.03 to 0.92).

Interpretation

For sparse tables, report Fisher's p-value rather than chi-squared. "Among drug-treated patients, 3 of 18 (17 %) experienced the event, compared to 12 of 32 (38 %) controls. Fisher's exact test gave OR 0.18 (95 % CI 0.03 to 0.92), p = 0.027."

Practical Tips

- Use Fisher's test when any expected cell count is below 5.
- The two-sided p-value uses R's default "minimum likelihood" rule; other software may use a different rule (doubled one-sided), giving slightly different values.
- The exact OR CI is not symmetric around the point estimate; report both bounds.
- For very large contingency tables, Fisher's exact test may be computationally prohibitive; use `simulate.p.value = TRUE` for Monte Carlo chi-squared instead.
- For paired / matched 2x2 data, use McNemar's exact test, not Fisher's.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Populations, samples, and the central limit theorem
- Bootstrap and permutation tests
- Maximum likelihood estimation
- One-sample t-test and one-proportion test
- Hypothesis testing, p-values, type I/II errors
- Two-sample and paired t-tests
- Two proportions, chi-square, risk and odds
- Pearson and Spearman correlation
- Non-parametric tests